Imperial College London | University of Sussex | UCL | UNEP WCMC

Projecting Responses of Ecological Diversity In Changing Terrestrial Systems (PREDICTS)

Predicts

Project funded by the Natural Environment Research Council (NERC).
Principal Investigator: Prof. Andy Purvis.

Data Entry Form – version PV07

Form compiled by: Joana Medeiros | Medeiros

Form completed on: 2026-07-16

Compiler's ID: JM1 *email:* j.medeiros@pgr.reading.ac.uk

I want to receive by email info on database progress: Unsure

Data source identification:

First author: N | Kingazi

Year of publication: 2 0 2 4 _

Publication/source title:
Tropical agroforestry supports insect pollinators and improves bean yield

If electronic copy of the publication submitted together with the form:

Publication type: Scientific article

Journal title: Journal of Applied Ecology

Data use permitted: For publication with proper referenc

A digital object identifier (DOI): 10.1111/1365-2664.14629

Link to data permission – reference from Insightly url:

EndNote reference: ırts insect pollinators and improves bean yield. Journal of A

Data provider or corresponding author contact details, if available:

Title | Nanyika Kingazi | Kingazi

Email: nanyikakingazi63@gmail.com

Tel:

Fax:

Skype:

Address:

Form compiler comments:
Coordinate evidence: Named study area matched in paper text/map evidence: Eastern lower slopes of Mount Kilimanjaro, northern Tanzania.; Species-count evidence: 30 species were found in agroforestry plots and 12 of them in; Site-count evidence: 16 sampling sites in one group plus 16 paired sampling sites in the comparison group = 32 total individual sites. Evidence: We examined 16

For the source ID generating, type 1st author / data provider's surname in the field below, ***USING STANDARD LATIN CHARACTERS ONLY:***

For NON-TYPICAL DATA SOURCE submitted together with the form – please attach zip-archive and provide file name here:

JM1_2024__Kingazi

Source ID: JM1_2024__Kingazi

Abstract (copy from the publication):

1. Pollination services contribute to crop productivity worldwide, but insect pollina- of Agriculture, Morogoro, Tanzania tors are declining in most agricultural landscapes, mainly due to agricultural ex- 2College of Forestry, Wildlife and Tourism, Sokoine University of Agriculture (SUA), pansion and an increase in intensive agricultural practices. To reduce the negative Morogoro, Tanzania effects of agricultural expansion and intensification, farmers can adopt ecologi- 3Tanzania Forestry Research Institute cal interventions, such as diversifying agricultural systems through agroforestry. (TAFORI), Morogoro, Tanzania 4Department of Ecology, Swedish However, there have been variable results on the effectiveness of agroforestry in University of Agricultural Sciences (SLU), enhancing pollinators and pollination services. Furthermore, most of the informa- Uppsala, Sweden tion has been generated in temperate regions, while the impact of agroforestry in 2. Using common beans (Phaseolus vulgaris L.) as the focal crop, this study tested whether a tropical agroforestry system called Chagga home gardens supports 3. The beans planted in the agroforestry plots had almost twice the abundance of insect pollinators, three times the richness of the species and almost twice the visitation rates than those planted in non- agroforestry plots. We also found a sig- nificant positive effect of woody cover in the surrounding landscape on insect pol- linator abundances, but not on species richness and visitation rates. Additionally, the abundance and richness of insect pollinators increased significantly with flower abundance, while the overall plant richness in a plot significantly increased insect pollinator abundance and their visitation rate. The difference in bean yield between unbagged flowers (in which insect pollinators were allowed to access bean flowers) and bagged flowers and the total yield were higher in agroforestry than in non- agroforestry plots. 4. Synthesis and applications. This study showed that, compared to monocul- ture, agroforestry generally promotes pollination services in a tropical context. Furthermore, we found that agroforestry is likely to be particularly helpful for © 2024 The Authors. Journal of Applied Ecology © 2024 British Ecological Society. | J Appl Ecol. 2024;61: 1067-1080. wileyonlinelibrary.com/journal/jpe 1067 1068 | KINGAZI et al. pollinators when it increases flower abundance and plant richness and if it is co- ordinated so that woody cover also increases at the landscape scale

Main Locations of data within the publication (page/s, table/s, figure/s):

Main PDF Docling tables: table_001.csv; table_002.csv; table_003.csv; Supplementary/downloaded files: Bean yield data.csv; Insect pollinator data.csv; NMDS_Data.csv; table_001.csv; table_002.csv; table_003.csv; page_001_ocr.txt; page_002_ocr.txt; page_003_ocr.txt; page_004_ocr.txt; page_005_ocr.txt; page_006_ocr.txt; page_007_ocr.txt; page_008_ocr.txt; page_009_ocr.txt; page_010_ocr.txt; page_011_ocr.txt; page_013_ocr.txt; page_014_ocr.txt; page_001_vision.json

If functional trait data are available, give file name/s below:

Part II – Study/ies data sheet. Source ID: JM1_2024__Kingazi

Study sequential number:	1	2	3	4	5
Study working title:	Tropical agrofores				
Study extent/coverage:	Eastern lower slo				
Location, as described by author:	Eastern lower slopes of Mount				
Continent / region:	Africa	- SELECT ONI	- SELECT ONI	- SELECT ONI	- SELECT ONI
Country/ies:	Tanzania	SELECT or ad	SELECT or ad	SELECT or ad	SELECT or ad
Sub-national unit/s or geographical name:	Eastern lower slo				
If study area map attached – record file name here:					
If GIS data provided by authors, attach zip-archive and type the file name here:					

Geographical coordinates for the approximate centre of each study area:

A. Latitude and Longitude in Decimal Degrees					
Latitude (Y, **negative for the southern hemisphere!***):*	-3.25				
Longitude (X, **negative for the western hemisphere!***):*	37.35				
B. Latitude and Longitude in Degrees, Minutes, Seconds ***(NOT REQUIRED IF coordinates in Decimal Degrees are available)***					
	deg. min. sec.	**deg. min. sec.**	**deg. min. sec.**	**deg. min. sec.**	**deg. min. sec.**
Latitude (dd/mm/ss and North or South):	S	X	X	X	X
Longitude (ddd/mm/ss and East or West):	E	X	X	X	X

Coordinates acquisition method:	Map & Google	Author's text	Author's text	Author's text	Author's text
Indication on location precision, as estimated by the form compiler:	ify agai Degre	N/A	N/A	N/A	N/A

Census or sampling method (uniform within a study):

Select census/sampling method or add new one:	Fixed plots/quad	Select or add ne	Select or add ne	Select or add ne	Select or add ne
Select abundance metric type or add new one:	abundance	Select or add ne	Select or add ne	Select or add ne	Select or add ne
Select abundance metric unit or add new:	individuals	Select or add ne	Select or add ne	Select or add ne	Select or add ne
Select sampling effort unit or add new:	plot	Select or add ne	Select or add ne	Select or add ne	Select or add ne

Period covered by the study and additional information:

Start (YYYY-MM-DD):	2022 03 01				
End (YYYY-MM-DD):	2022 05 31				
Number of sites within the study:	32				
Number of species, taxa or groups of species sampled:	30				
	Entire commur	Entire commur	Entire commur	Entire commur	Entire commur
Compiler's comments:					

Take-down Policy.

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